A Project Stage-I Report On

DETECTION AND RECOMMENDATIONS IN PEANUT FARMING

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

from

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

By

BHUVANESHWARI REBBA (22C11A0525)

AKANKSHA KANDIBANDA (22C11A0509)

Under the guidance of

Dr. J. Srinivas Rao **M.Tech, Ph.D**

Associate Professor

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ANURAG ENGINEERING COLLEGE

(An Autonomous Institution)

(Affiliated to JNTUH, Hyderabad & Approved by AICTE, New Delhi)

Ananthagiri (V & M), Kodad, Suryapet (Dt.), Telangana -508206.

2025 – 26

ANURAG ENGINEERING COLLEGE

(An Autonomous Institution)

(Affiliated to JNTUH, Hyderabad & Approved by AICTE, New Delhi)
Ananthagiri (V & M), Kodad, Suryapet (Dt), Telangana -508206.

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

*This is to certify that the project work entitled "**DETECTION AND RECOMMENDATIONS IN PEANUT FARMING**" is a bonafide work done by "**BHUVANESHWARI REBBA (22C11A0525) and AKANKSHA KANDIBANDA (22C11A0509),**" in the partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering from JNTUH, Hyderabad during the year **2025-26**.*

This work has not been submitted to any other university or institute or organization for the award of any degree or diploma.

(Dr. J. Srinivas Rao)
Supervisor

(Dr. Y.V.R. Naga Pawan)
H.O.D.

DECLARATION

We, the undersigned IV B.Tech CSE I Semester students, hereby declare that the Project Stage-I report titled **"Detection and Recommendation in Peanut Farming"** is our original work carried out under the guidance of **Dr. J. Srinivas Rao Ph.D, Associate Professor, Department of Computer Science and Engineering** at **ANURAG Engineering College, Ananthagiri (V&M), Kodad, Suryapet (Dt.), Telangana – 508206**.

This report has not been submitted elsewhere. All sources are cited in **References**. We agree with the college academic integrity.

BHUVANESHWARI REBBA
(22C11A0525)

Date: 24-11-2025
Place: Ananthagiri (V&M)

AKANKSHA KANDIBANDA
(22C11A0509)

ACKNOWLEDGEMENTS

This report will certainly not be completed without due acknowledgement paid to all those who helped me during this project work.

We express sincere thanks to my supervisor **Dr. J. Srinivas Rao,** for giving moral support, kind attention and valuable guidance throughout this project work.

It is our privilege to thank **Dr. Y. V. R. Naga Pawan**, Head of the Department, Department of Computer Science and Engineering for his encouragement during the progress of this project work.

We are thankful to both Teaching and non-teaching staff members of the Department **of Computer Science and Engineering** for their kind cooperation and all sorts of support in bringing out this project work successfully.

We derive great pleasure in expressing my sincere gratitude to the Principal **Dr. T. Suresh Kumar** for his timely suggestions, which helped us to complete this work successfully.

We also thank the **Management of ANURAG Engineering College** for providing the necessary facilities.

Finally, we are indebted to **our parents and friends** for their constant support and encouragement.

BHUVANESHWARI REBBA

(22C11A0525)

AKANKSHA KANDIBANDA

(22C11A0509)

CONTENTS

Abstract		i
List Of Figures		ii
List Of Tables		iii
List Of Acronyms		iv
1	Introduction	1
2	Objectives	2
3	Literature Survey	3-6
4	Existing System	7
5	Problem Statement	8
6	Proposed System	9
7	System Architecture	10-12
8	Technologies Used	13-14
9	Software Requirements	15
10	Hardware Requirements	16
11	Front End Requirements	17
12	Back End Requirements	18
13	Modules	19
14	System Design	20-24
References		25

Abstract

The project “Detection and Recommendations in Peanut Farming” focuses on detecting pests, diseases, and crop damage in peanut plants using machine learning and deep learning algorithms. It analyzes various plant parts such as stems, pods, roots, leaves, and insects to provide complete crop health assessment. The system employs the YOLOv8 deep learning model for accurate image-based detection of pests and visible symptoms, while a Random Forest algorithm is used for classification and prediction of disease severity. The approach combines visual analysis and predictive modeling to identify affected areas and recommend suitable pesticides. This intelligent system helps farmers detect issues early, minimize crop loss, and make data-driven decisions to improve productivity and sustainability in peanut cultivation.

List Of Figures

Figure No.	Description	Page No.
7.1	System Architecture	10
14.1	Use Case Diagram	20
14.2	Activity Diagram	21
14.3	Class Diagram	22
14.4	Sequence Diagram	23
14.5	Collaboration Diagram	24

List Of Tables

Table No.	Description	Page No.
3.1	Literature Survey	3

List Of Acronyms

AI	:	Artificial Intelligence
API	:	Application Programming Interface
DB	:	Database
ML	:	Machine Learning
RF	:	Random Forest
UI	:	User Interface
YOLOv8	:	You Only Look Once – Version 8

CHAPTER 1
INTRODUCTION

Peanut cultivation is a vital part of agriculture, contributing significantly to food production, edible oil manufacturing, and rural income. However, peanut crops are frequently affected by pests and diseases that damage various plant parts such as stems, pods, roots, leaves, and seeds. These infestations often go unnoticed in the early stages, resulting in severe yield loss and economic impact. Traditional methods of crop monitoring rely on manual inspection, which is time-consuming, labor-intensive, and prone to human error. Therefore, an automated detection system is essential to assist farmers in monitoring crop health effectively and taking timely action.

The project "Detection and Recommendations in Peanut Farming" focuses on developing an intelligent and automated system capable of detecting and classifying pests, diseases, and crop damage using machine learning and deep learning techniques. The system processes images of peanut plants to identify the affected areas and determine the type and severity of damage. The YOLOv8 deep learning model is employed for object detection and localization of visible pests or symptoms, while the Random Forest algorithm is utilized for classification and prediction based on extracted features.

By combining these two techniques, the system ensures high accuracy and reliability in identifying crop issues. Once the problem is detected, it suggests suitable pesticide recommendations for the specific pest or disease. This integrated approach enables farmers to make quick, informed decisions and adopt preventive measures to protect their crops.

Overall, the project highlights the effective use of machine learning and image-based analysis in agriculture to improve pest management, reduce losses, and enhance the productivity of peanut farming.

CHAPTER 2
OBJECTIVES

2.1 OBJECTIVES:

- To develop an automated image-based system that detects and classifies pests, diseases, and crop damage across multiple peanut plant parts such as stems, pods, roots, leaves, and insects.
- To implement machine learning algorithms — YOLOv8 for detection and Random Forest for classification — ensuring high accuracy and reliability.
- To recommend appropriate pesticides based on the identified pest or disease for effective crop management.
- To provide a fast, efficient, and user-friendly platform that assists farmers in making informed decisions, reducing yield loss, and improving peanut crop productivity.

CHAPTER 3
LITERATURE SURVEY

3.1 Literature Survey

The study titled **"**Online Recognition of Peanut Leaf Diseases Using Data Balance Algorithm and Deep Transfer Learning" (2023) by Q. Feng et al. introduces a transfer-learning-based approach to peanut leaf disease identification. The authors highlight that dataset imbalance is a major issue in agricultural image classification, as certain diseases are under-represented in real farm environments. To address this, the paper integrates a data balancing algorithm with deep transfer learning models, improving recognition accuracy across diverse lighting and background conditions. Their findings demonstrate enhanced robustness and reliability in disease detection, even in complex field scenarios. This research provides valuable insights for our project, particularly in terms of adopting transfer learning strategies and implementing data balancing techniques to improve the detection accuracy and consistency of peanut pest and disease classification.

3.2 Literature Survey

The research paper "Hyperspectral Technique for Detection of Peanut Leaf Spot Disease Based on Improved PCA Loading" (2023) by Q. Guan et al. utilizes hyperspectral imaging combined with an enhanced PCA (Principal Component Analysis) loading method to identify peanut leaf spot disease. The improved PCA approach significantly reduces noise while preserving highly discriminative spectral features, allowing more precise classification of disease symptoms. Hyperspectral data offer rich spectral signatures that traditional RGB images cannot capture, making the technique particularly suitable for early disease detection. This study demonstrates the effectiveness of dimensionality-reduction techniques in optimizing hyperspectral datasets. The concepts from this work can be incorporated into our system to improve feature extraction efficiency and enhance model performance in detecting peanut crop abnormalities.

3.3 Literature Survey

The paper "Detection of Peanut Leaf Spot Disease Based on Leaf-, Plant-, and Field-Scale Hyperspectral Reflectance" (2022) by Q. Guan, D. Zhao, and colleagues presents a multi-scale investigation of peanut leaf spot detection using hyperspectral reflectance data. Their methodology includes capturing spectral information at the leaf, whole-plant, and field levels, enabling a comprehensive understanding of disease progression under real agricultural conditions. The authors report that hyperspectral signatures change noticeably with disease severity, making it a powerful tool for early and accurate detection. This multi-level analysis ensures high reliability and adaptability to outdoor environments. Insights from this research can enhance our project by enabling more accurate diagnosis of disease severity and improving early detection capabilities beyond simple image-based methods.

3.4 Literature Survey

The study "Detection of Peanut Leaf Spots Disease Using Canopy Hyperspectral Reflectance" (2019) by T. Chen et al. focuses on detecting peanut leaf spot diseases through canopy-level hyperspectral measurements. Unlike pure leaf-level studies, canopy reflectance captures the overall plant spectral response, allowing earlier detection before visible symptoms appear. The researchers demonstrate that spectral variations associated with disease onset can be detected even at low infection levels, providing farmers with valuable early warning signals. Their findings emphasize the importance of remote sensing techniques in large-scale crop monitoring. Incorporating these concepts into our project can significantly improve the early-stage detection of peanut pests and diseases, enabling timely interventions and reducing crop loss.

3.5 Literature Survey

The foundational paper "Using Deep Learning for Image-Based Plant Disease Detection" (2016) by S. P. Mohanty, D. P. Hughes, and M. Salathé introduces the use of deep convolutional neural networks (CNNs) for plant disease recognition. The authors demonstrate

that deep learning models outperform traditional machine learning techniques in feature extraction and classification accuracy. Their work sets the groundwork for modern agricultural AI systems by proving the effectiveness of end-to-end learning from large image datasets. The concepts in this research—particularly CNN-based feature extraction and automated image recognition—are crucial for improving detection accuracy in our peanut farming system. This study supports the selection of advanced deep learning frameworks such as YOLOv8 for real-time pest and disease detection.

S.No	Title of the Paper	Authors & Year	Methods / Techniques Used	Key Findings	Relevance to Current Project
1	Online Recognition of Peanut Leaf Diseases Using Data Balance Algorithm and Deep Transfer Learning	Q. Feng et al., 2023	Transfer Learning, Data Balancing Algorithm	Improved disease recognition accuracy in imbalanced and diverse farm environments	Helps improve robustness of peanut disease detection using balanced datasets and transfer learning
2	Hyperspectral Technique for Detection of Peanut Leaf Spot Disease Based on Improved PCA Loading	Q. Guan et al., 2023	Hyperspectral Imaging, Improved PCA Loading	Noise reduction and extraction of discriminative spectral features	Provides idea for PCA-based dimensionality reduction to enhance feature extraction in our system
3	Detection of Peanut Leaf Spots Disease Using Canopy Hyperspectral Reflectance	T. Chen et al., 2019	Canopy Hyperspectral Reflectance	Enables early detection before visible symptoms appear	Helps implement early-warning systems for peanut crop diseases
4	Using Deep Learning for Image-Based Plant Disease Detection	S.P. Mohanty et al., 2016	Convolutional Neural Networks (CNNs)	CNNs outperform traditional ML methods for plant image classification	Forms the foundation for using YOLOv8 and deep learning in peanut disease detection

Fig 3.1 Literature Survey

CHAPTER 4
EXISTING SYSTEM

In the existing agricultural systems, pest and disease detection in peanut crops is primarily based on manual field inspection and basic image processing techniques, where farmers or experts visually identify symptoms such as leaf spots, pest damage, or discoloration. Some recent studies, like those using CNN, MFO, and EViTA algorithms, have improved accuracy through automated image classification and feature selection; however, these models are often limited to leaf-level analysis and depend on controlled environmental conditions. They lack the ability to analyze multiple plant parts such as stems, pods, roots, and insects, and do not include pesticide recommendation mechanisms. Furthermore, high computational requirements and the absence of real-time detection reduce their effectiveness for on-field use, making these systems less adaptable for practical peanut farming applications.

4.1 DRAWBACKS

- **Limited Scope:** Focuses mainly on leaf images; ignores stems, pods, roots, and insect-level detection.
- **No Recommendation System:** Existing models detect pests but do not provide pesticide or treatment suggestions.
- **Low Adaptability:** Performance decreases under natural lighting or complex field backgrounds.
- **Dataset Limitation:** Peanut-specific datasets are limited, reducing model generalization to diverse farm conditions.

CHAPTER 5
PROBLEM STATEMENT

Pest and disease detection in peanut crops is often performed through manual observation or simple image-based methods, resulting in low accuracy and slow diagnosis. Deep learning models such as CNN combined with MFO and EViTA have improved feature extraction and classification, but their application remains limited to leaf-level analysis under controlled conditions. These approaches lack the ability to detect issues in stems, pods, roots, and insects, and do not provide pesticide recommendations. The absence of a comprehensive, field-adaptable solution restricts their effectiveness in practical peanut farming environments.

CHAPTER 6
PROPOSED SYSTEM

The proposed system uses machine learning and deep learning techniques to automatically detect pests, diseases, and crop damage in peanut plants. It processes images of different plant parts such as stems, pods, roots, leaves, and insects using the YOLOv8 model for real-time object detection and feature extraction. The detected results are further classified by a Random Forest algorithm, which predicts the type and severity of the pest or disease. Based on the classification, the system provides suitable pesticide recommendations from an integrated database. Implemented in Python, this system ensures high accuracy, faster processing, and reliable decision-making to assist farmers in effectively monitoring crop health and managing pest-related issues.

6.1 ADVANTAGES:

- **Automation:** Eliminates manual inspection through automated image analysis and prediction.
- **Comprehensive Analysis:** Supports detection across multiple plant parts—stems, pods, roots, leaves, and insects.
- **User-Friendly Interface:** Provides an efficient and easy-to-use interface for farmers and agricultural users.
- **Scalability:** Can be extended to other crops and integrated with larger agricultural monitoring systems.
- **Improved Reliability:** Reduces human error and ensures consistent detection and recommendation results.

CHAPTER 7
SYSTEM ARCHITECTURE

7.1 SYSTEM ARCHITECTURE DIAGRAM:

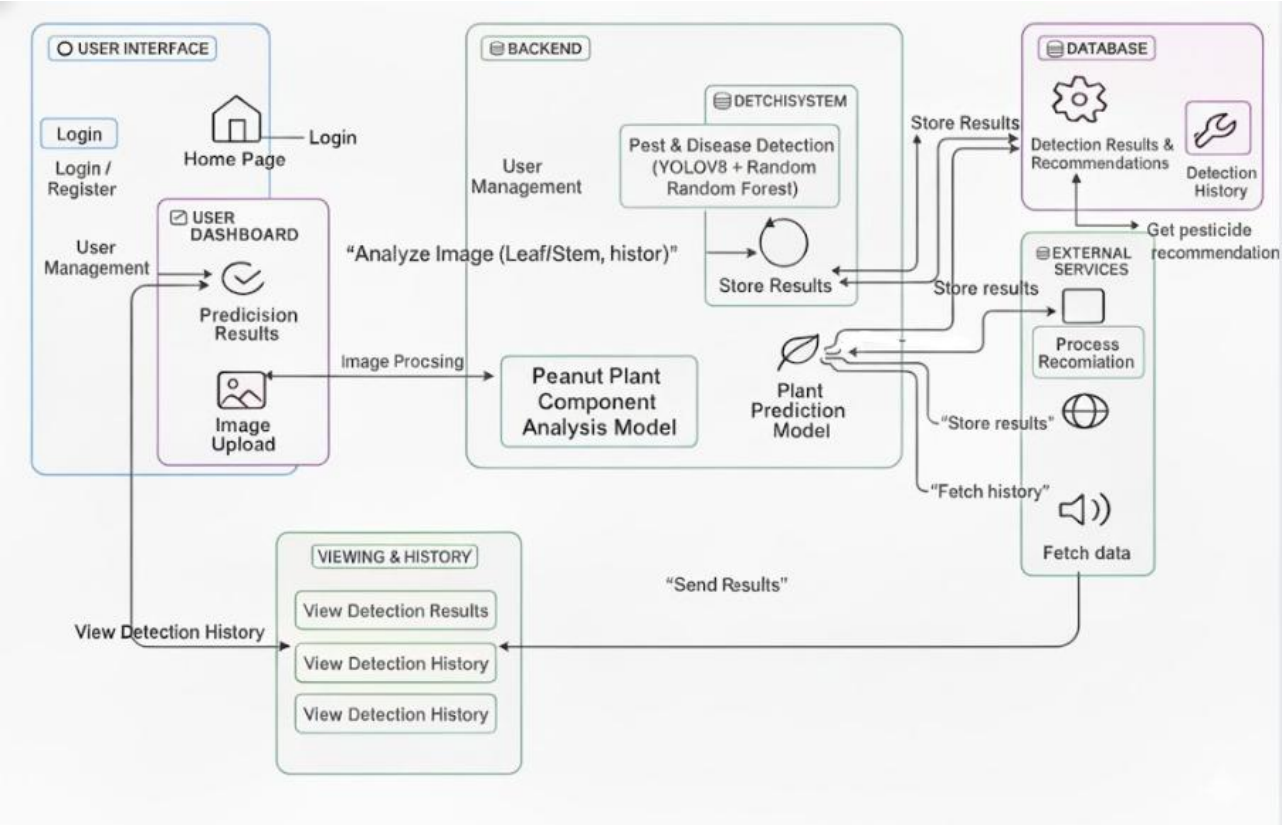

Fig 7.1 System Architecture

7.1.1 USER:

The user or admin interacts with the system through a web interface. Farmers or agricultural officers can log in and upload images of peanut plant parts such as leaves, stems, pods, roots, or insects for detection and diagnosis.

7.1.2 UPLOAD IMAGE:

The uploaded images are accepted through the system interface. Each image represents a part of the peanut plant suspected to have pest infestation or damage. The image data serves as the input for the detection pipeline.

7.1.3 IMAGE PREPROCESSING:

Before analysis, images are pre-processed to improve detection accuracy. This includes operations such as resizing, noise removal, normalization, and background adjustment to ensure compatibility with the YOLOv8 model.

7.1.4 YOLOv8 DETECTION:

The pre-processed images are passed into the YOLOv8 deep learning model, which performs real-time detection of pests, diseases, and damage. The model identifies objects in the image and classifies them into categories such as leaf damage, pod rot, insect infestation, or stem infection.

7.1.5 RANDOM FOREST CLASSIFICATION:

The detected regions are then analyzed by the Random Forest classifier, which refines predictions by learning patterns from various features (color, shape, texture, etc.). It classifies the pest or disease type and predicts severity levels.

7.1.6 PESTICIDE RECOMMENDATION DATABASE:

Once the pest or disease type is identified, the system retrieves corresponding pesticide or treatment recommendations from a pre-built database. This includes information such as pesticide name, description, usage instructions, and image for reference.

7.1.7 RESULT DISPLAY:

Finally, the system presents the detection results to the user. The display includes the

detected pest or damage, confidence score, severity level, and recommended pesticide. This helps farmers take quick, data-driven action to protect their crops.

CHAPTER 8
TECHNOLOGIES USED

8.1 PYTHON:

Python is the core programming language used for model development, backend integration, and data processing. Its vast libraries make it ideal for implementing machine learning and deep learning workflows efficiently.

8.2 FLASK:

Flask is a lightweight Python web framework used to build the backend API. It connects the YOLOv8 and Random Forest models with the React frontend, handling image uploads, model predictions, and pesticide data retrieval.

8.3 YOLOv8 (You Only look Once v8):

YOLOv8 is a state-of-the-art deep learning model used for real-time pest and disease detection. It accurately identifies and localizes affected parts such as leaves, stems, pods, roots, and insects from uploaded images.

8.4 RANDOM FOREST:

Random Forest is a supervised machine learning algorithm implemented using Scikit-learn.

It classifies detected regions and predicts pest type and severity level, improving decision accuracy and robustness.

8.5 REACT.JS:

React.js is used to develop the **frontend user interface**. It enables users to upload images, view detection results, and get pesticide recommendations dynamically with a responsive design.

8.6 MONGODB:

MongoDB is a NoSQL database used to store pesticide details, user history, and detection outputs. It ensures fast data access and scalability for real-time applications.

8.7 SUPABASE:

Supabase provides authentication and database integration services for secure user login, signup, and cloud storage of detection history.

8.8 AXIOS:

Axios is used in the React frontend for communication with the Flask backend API. It handles requests for pest detection and pesticide recommendation seamlessly.

8.9 JOBLIB:

Joblib is used to save and load the trained Random Forest model efficiently. This allows the model to be reused without retraining, improving performance during deployment.

CHAPTER 9
SOFTWARE REQUIREMENTS

9.1 SOFTWARE REQUIREMENTS:

- Operating System : Windows 10 and Above
- Programming Language : Python 3.9 and Above
- Tool used : VS-Code
- Library used : Random Forest

CHAPTER 10
HARDWARE REQUIREMENTS

10.1 HARDWARE REQUIREMENTS:

Processor	:	Minimum Intel i5
RAM	:	Minimum 4 GB
Storage	:	Minimum 512 GB

CHAPTER 11
FRONT END REQUIREMENTS

11.1 REACT.JS:

React.js is a JavaScript library used to create a dynamic, component-based, and interactive user interface. In this system, it allows users to upload peanut plant images, view pest detection results, and receive pesticide recommendations in real time. React efficiently manages UI updates, ensuring smooth and responsive interactions.

11.2 TAILWIND CSS:

Tailwind CSS is used for designing the user interface. It provides a modern, clean, and mobile-friendly layout, making the application visually appealing and easy to navigate. It helps maintain consistent styling across all components without writing lengthy CSS code.

11.3 AXIOS:

Axios is a JavaScript library used for making HTTP requests between the frontend and backend. It enables seamless communication with the Flask API for tasks such as sending uploaded images and fetching detection and pesticide recommendation results.

11.4 VITE:

Vite is a fast frontend build tool used to run and compile the React project efficiently. It ensures faster development, hot-reloading, and quick bundling, improving the overall performance of the frontend during development and deployment.

CHAPTER 12
BACKEND REQUIREMENTS

12.1 PYTHON:

Python is the main backend programming language that integrates all logic, including image analysis, model training, and prediction. It runs the YOLOv8 and Random Forest models and manages communication between the backend and frontend.

12.2 MONGODB:

MongoDB is a NoSQL database used to store pesticide data, detection history, and user information. It ensures quick data access and scalability for real-time operations.

12.3 SUPABASE:

Supabase is used for authentication and cloud database connectivity. It provides secure access control and stores user-related data.

CHAPTER 13
MODULES

13.1 IMAGE UPLOAD AND PREPROCESSING MODULE:

This module allows users to upload peanut plant images through the web interface. The uploaded image may include parts such as leaves, stems, roots, pods, or insects. Once uploaded, the system performs preprocessing operations like resizing, normalization, and format conversion to ensure compatibility with the detection model and maintain result accuracy.

13.2 DETECTION AND CLASSIFICATION MODULE: (YOLOv8+RANDOM FOREST)

This is the core module of the system that combines YOLOv8 and Random Forest algorithms. YOLOv8 is used for detecting pests, diseases, and damaged crop regions in real time, providing bounding boxes and confidence levels for each identified area. The Random Forest classifier then analyzes the detected regions to classify pest type and severity level. Together, these models provide a powerful hybrid approach, ensuring accurate and efficient detection and classification in peanut crops.

13.3 PESTICIDE RECOMMENDATION MODULE:

Based on the detected pest type and severity, this module retrieves suitable pesticide information from the MongoDB database. It displays pesticide names, descriptions, and usage instructions, helping farmers make informed and timely decisions for pest management.

13.4 RESULT MODULE:

This module handles user authentication, login, and history management through Supabase. It securely stores detection records, past results, and pesticide recommendations. Users can access their previous analyses and view detailed reports directly from the web dashboard.

CHAPTER 14
SYSTEM DESIGN

14.1 USE CASE DIAGRAM:

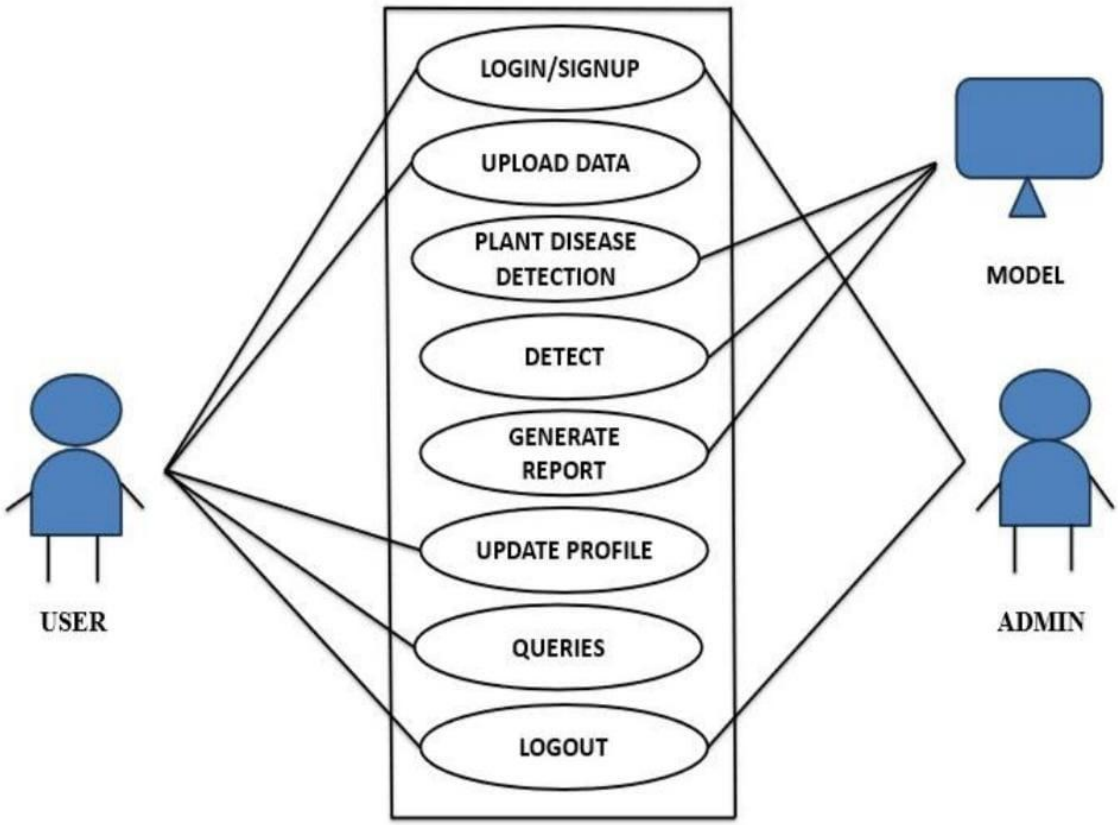

Fig 14.1 Use Case Diagram

14.2 ACTIVITY DIAGRAM:

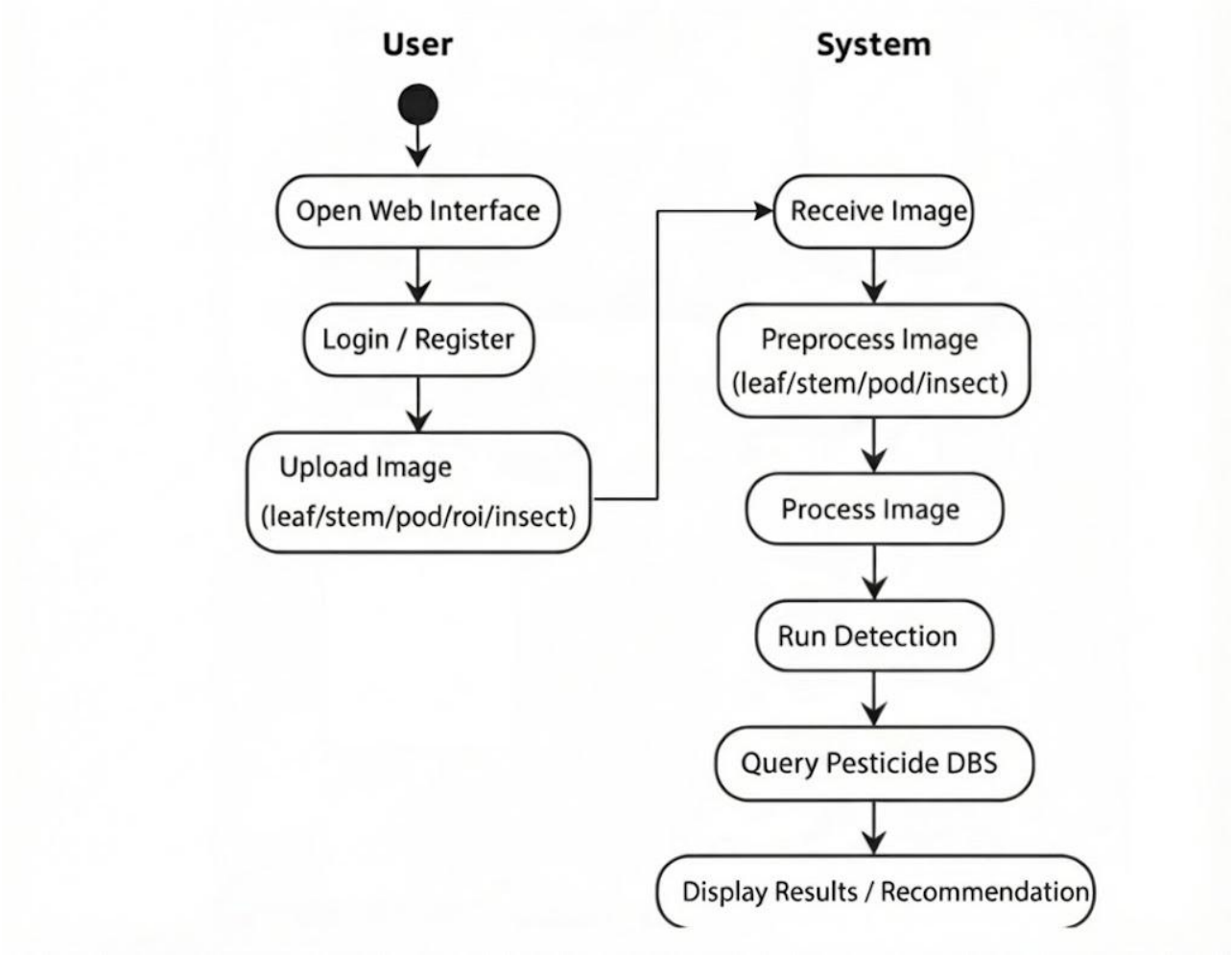

Fig 14.2 Activity Diagram

14.3 CLASS DIAGRAM:

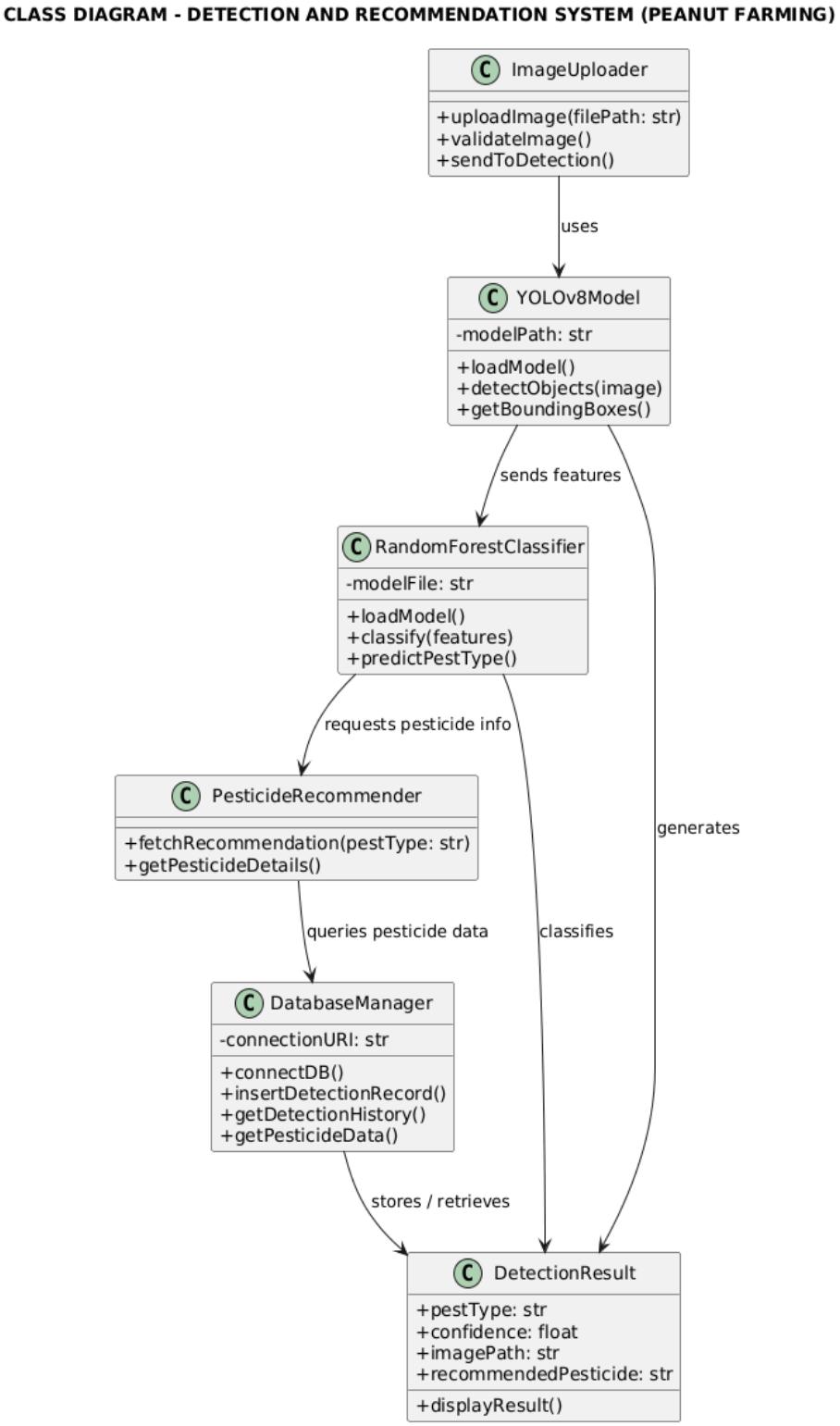

Fig 14.3 Class Diagram

14.4 SEQUENCE DIAGRAM:

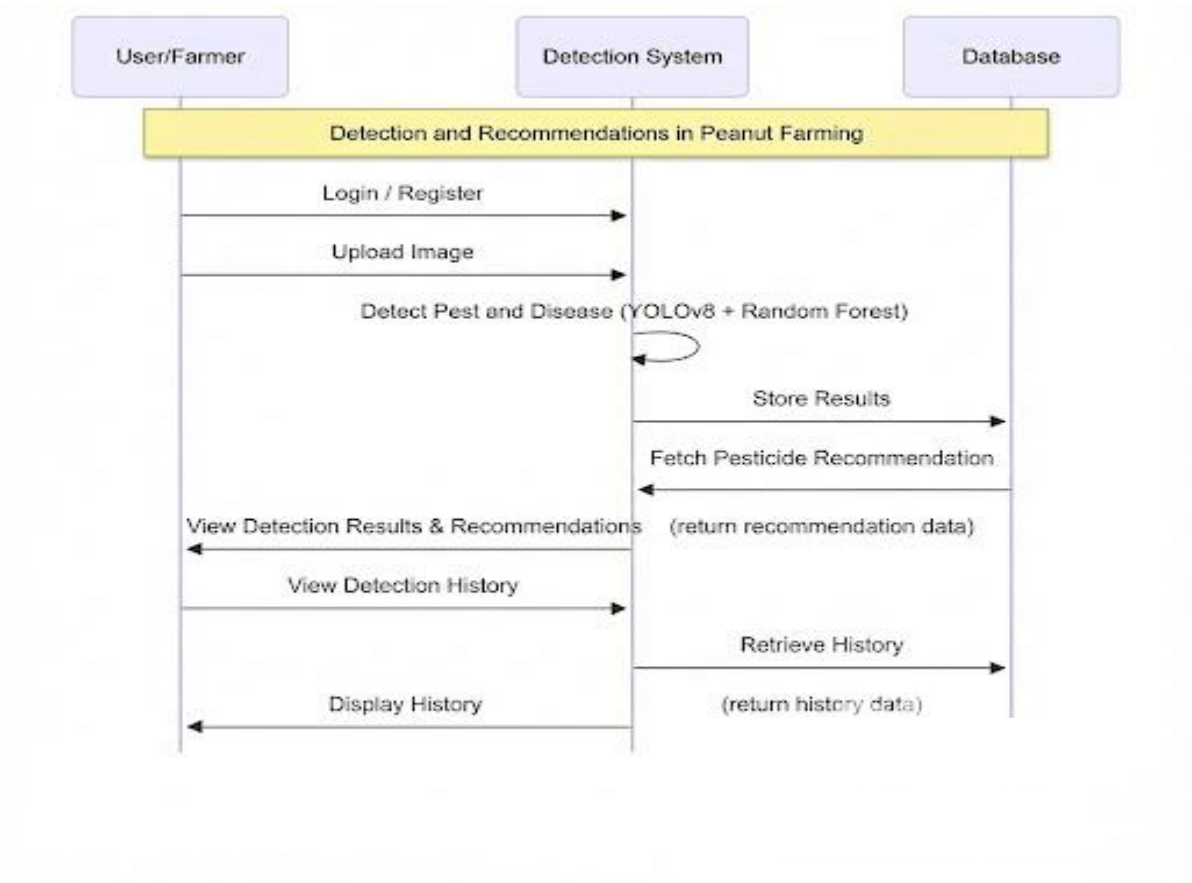

Fig 14.4 Sequence Diagram

14.5 COLLABORATION DIAGRAM:

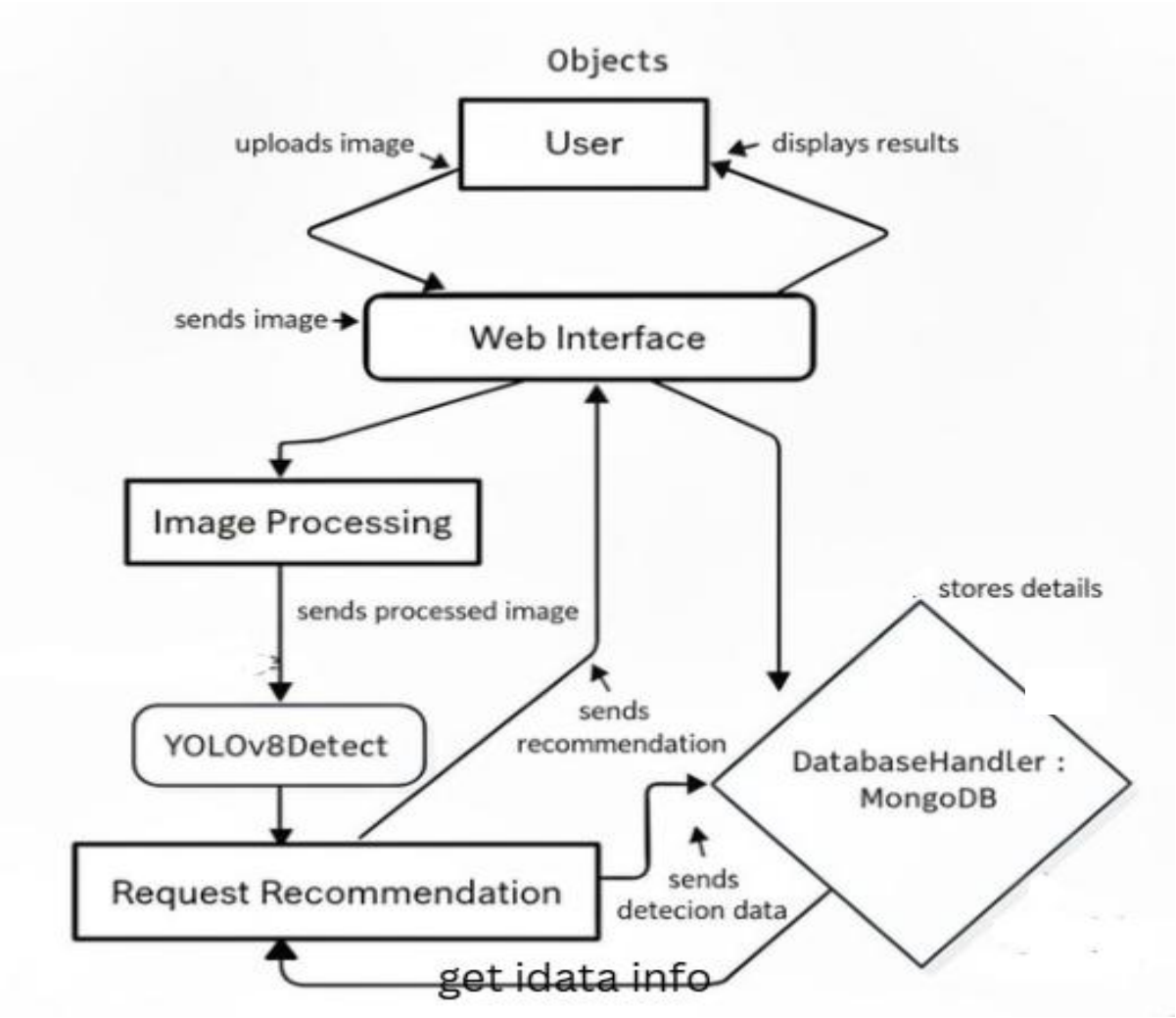

Fig 14.5 Collaboration Diagram

REFERENCES

- Feng, Q.; Xu, P.; Ma, D.; Lan, G.; Wang, F.; Wang, D.; Yun, Y. "Online Recognition of Peanut Leaf Diseases Based on the Data Balance Algorithm and Deep Transfer Learning." *Precision Agriculture*, 2023, 24, 560–586.
- Guan, Q.; Zhao, D.; Feng, S.; Xu, T.; Wang, H.; Song, K. "Hyperspectral Technique for Detection of Peanut Leaf Spot Disease Based on Improved PCA Loading." *Agronomy*, 2023, 13, 1153.
- Guan, Q.; Song, K.; Feng, S.; Yu, F.; Xu, T. "Detection of Peanut Leaf Spot Disease Based on Leaf-, Plant-, and Field-Scale Hyperspectral Reflectance." *Remote Sensing*, 2022, 14, 4988.
- Chen, T.; Zhang, J.; Chen, Y.; Wan, S.; Zhang, L. "Detection of Peanut Leaf Spots Disease Using Canopy Hyperspectral Reflectance." *Computers and Electronics in Agriculture*, 2019, 156, 677–683.
- Mohanty, S.P.; Hughes, D.P.; Salathé, M. "Using Deep Learning for Image-Based Plant Disease Detection." *Frontiers in Plant Science*, 2016, 7, 1419.
- Shahi, T.B.; Xu, C.-Y.; Neupane, A.; Fresser, D.; O'Connor, D.; Wright, G.; Guo, W. "A Cooperative Scheme for Late Leaf Spot Estimation in Peanut Using UAV Multispectral Images." *PLOS ONE*, 2023, e0282486.
- Pinto, J.; Powell, S.; Peterson, R.; Rosalen, D.; Fernandes, O. "Detection of Defoliation Injury in Peanut with Hyperspectral Proximal Remote Sensing." *Remote Sensing*, 2020, 12, 3828.
- Guclu, V.; Aydogdu, M.; Basak, M.; Kizil, S.; Uzun, B.; Yol, E. "Characterization of a Groundnut Collection to Stem Rot Disease Caused by *Sclerotium rolfsii*." *Australasian Plant Pathology*, 2020, 49, 691–700.
- Y.; Liao, J.; He, Z.; Khurshid, M.; Wang, C.; Zhang, Z.; Mao, J.; Xia, Y. "Effects of Peanut Rust Disease on Agricultural Production: Current Control Strategies and Progress in Breeding for Resistance." *Genes*, 2024, 15, 102.